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4) How long does it take a packet of length 1,000 bytes to propogate over a link of distance 2,500 km, propagation speed $2.5 * 10^8$ m/s, and transmission rate 2 Mbps?

Does this delay depend on packet length? Does this delay depend on transmission rate?

1 ⇒ a) calculate Transmission Delay

Transmission delay (trans) is the time required to push all the packet's bits onto the link

Step ①

we can only send data in bits

$L = 1000$ bytes $\Rightarrow 8 \times 1000 = 8000$ bits

$R = 2$ Mbps $= 2 \times 10^6$ bps

$d = 2500$ km

$S = 2.5 \times 10^8$ m/s.

step ② Calculate Transmission delay

$$\boxed{d_{(trans)} = \frac{L}{R}}$$

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(f) The first bit has reached Host B.

(g) Suppose $S = 2.5 \times 10^8$, $L = 1500$ bytes & $R = 10$ Mbps. Find the distance that d_{prop} equals d_{trans}

$\Rightarrow$ $S = 2.5 \times 10^8$

$L = 1500 \times 8 \Rightarrow 12000$ bits

$R = 10 \times 10^6$ bps.

$m = ?$

$$d_{prop} = d_{trans}$$

$$\frac{d}{S} = \frac{L}{R} \Rightarrow \frac{m}{S} = \frac{L}{R}$$

$$m = \frac{L}{R} \times S \Rightarrow \frac{1200\cancel{0}}{1\cancel{0} \times 1\cancel{0}^{\cancel{8}}} \times 2.5 \times 10^{8}$$

$$= 1200 \times 2.5 \times 10^2 \Rightarrow 300 \text{ km}$$

(h) For the same question calculate d_{trans} (L/R

$$= \frac{12000}{10 \times 10^6} \Rightarrow \frac{12 \times 10^{\cancel{3}}}{10^{\cancel{7}\,4}} = 12 \times 10^{-4}$$

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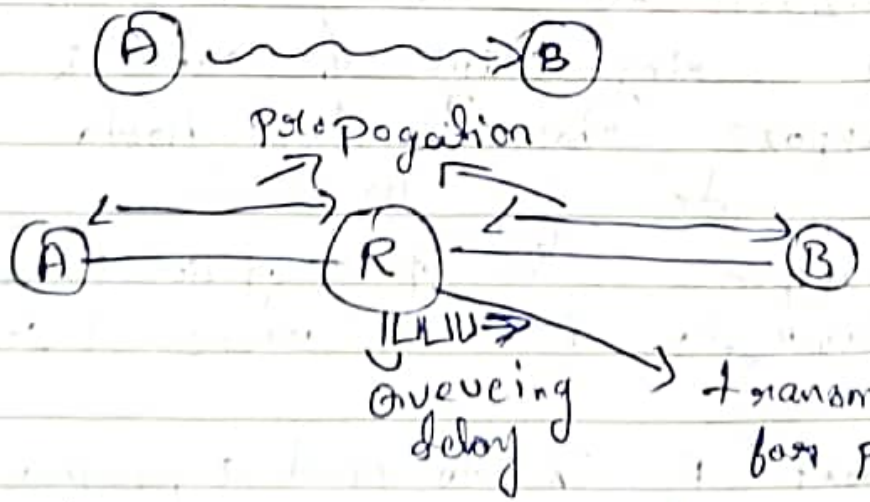
a) Express the propogation delay, d_{prop}, in terms of m & s.

b) Determine the transmission time of the packet, d_{trans} in terms of L & R.

c) Ignoring processing and queuing delay obtain an expression for the end-to-end delay.

d) Suppose Host A begins to transmit the packet at time $t = 0$.
At time $t = d_{trans}$, where is the last bit of the packet?

e) Suppose d_{prop} is greater than d_{trans}. At time $t = d_{trans}$, where is the first bit of packet?



(f) if d_{prop} is less than d_{trans} At time $t = d_{trans}$ " 1st packet

① Consider sending a packet from a sou host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays are constant and which are variable?

Ans: The delay components are processing delays, transmission delay, propagation delays and queuing delay. All of these delays are fixed, except for the queuing delays, which are variable.

② Suppose Host A wants to send a large file to Host B. The path from Hos A to Host B has three links, of rates R1 = 500 kbps, R2 = 2 Mbps and R3 = 1 Mbps.

b. Suppose the file is 4 million bytes (4 Mb.) Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B?

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1 MB = 1 MegaBytes
↓
1 Megabites

Decimal Method

1MB = 1 x 1000000 Bytes

= 1 x 1000000 x 8 bites

1MB = 8000000 bites

Binary Method

1MB = 1024 kB → 1024 x 1024

= 1,048,576 Bytes

= 8 x 1,048,576

1MB = 8,388,608 bites

Decimal Method

1MB = 1×10^6 Bytes

= $1 \times 10^3 \times 10^3$ Bytes

1MB = 1000 kB

1KB = 1000 Bytes

1KB = 8000 bites

500 kB = ?

500 kB = 4,000,000 bites

Binary method

1 KB = 1024 Bytes

~~500 KB = 512,00~~

1KB = 8192 bites

500 KB = 4,096,000

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$$d_{trans} = \frac{8000}{2 \times 10^6} = 0.004 \text{ sec}$$

$$= 4 \times 10^{-3} \text{ sec}$$

$$d_{trans} = 4 \text{ ms}$$

if L is 4000 bytes $\Rightarrow 4000 \times 8$

$= 32000$ bits

$$d_{trans} = \frac{32000}{2000000} = 0.016$$

$$= 16 \text{ ms}$$

Step (3) Propogation delay (d_{prop}) is the time it takes for a single bit to travel across the distance of the link. It is independent of the packet length & transmission rate.

$d = 2500$ km $\Rightarrow 2500 \times 10^3$ m

propogation speed (s) $= 2.5 \times 10^8$

$$d_{(prop)} = \frac{d}{s} \Rightarrow \frac{2500 \times 10^3}{2.5 \times 10^8}$$

$$= 1000 \times 10^{-5}$$

$$= 0.01 \text{ sec}$$

$$= 10 \text{ ms}$$

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c. throughput = {50 kbps, 100 kbps, 1 Mbps}

time taken to transfer = $\frac{32000000}{100000}$ = 320 sec

d. time taken to transfer = $\frac{32000000}{1000000}$ = 32 sec

throughput = {5 Mbps, 10 Mbps, 1 Mbps}

(3) This elementary problem begins to explore propogation delay and transmission delay, two central concepts in data networking.

Consider two hosts A & B, connected by a single link of rate R bps. Suppose that the two hosts are separated by m meters, and suppose the propagation speed along the link is s meters/sec.

Host A is to send a packet of size L bits to Host B.

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$d_{end-to-end} = 8.192 + 0.004 \Rightarrow 8.19$

$L = 1Mb = 1 \times 10^6$ bytes

$= 1 \times 8 \times 10^6$ bits

$= 8 \times 10^6$ bits

$= 8000000$

$d_{trans} = \frac{L}{R} = \frac{8000000}{1000000} = 8$ sec

$= 8000$ ms

$d_{prop} = 0.004$ ms

$d_{end-to-end} = 8000 + 0.004 \Rightarrow 8000.004$

$= 8.000004$ sec

1 byte = 8 bits

1 Nibble = 4 bits

1 Kb = 1024 byte = 8192 bits

1 Mb = 1024 Kb = $1024 \times 1024 \times 8$

$= 8,388,608$ bits

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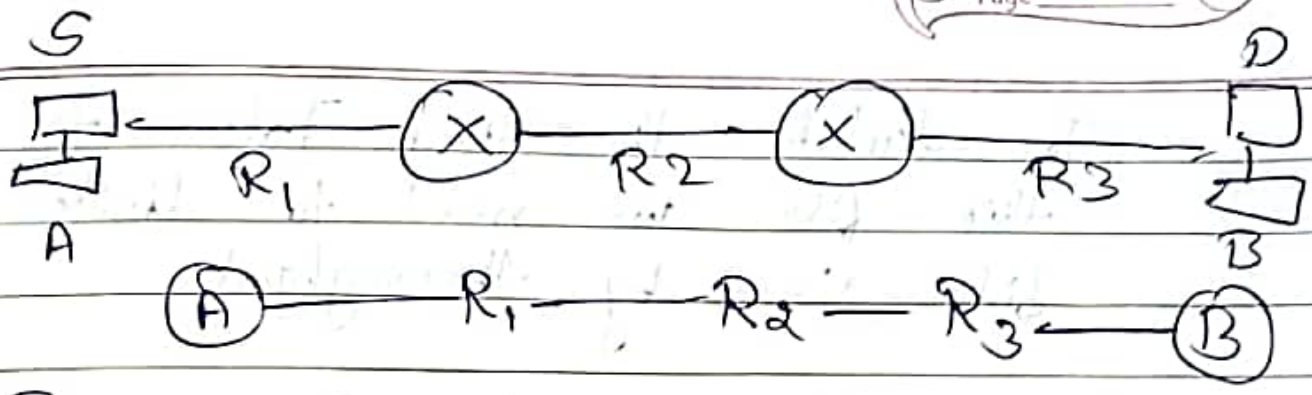

$R1 = 500$ Kbps
$R2 = 2$ Mbps
$R3 = 1$ Mbps

file size $= 4 \times 10^6$ bytes
$= 4 \times 8 \times 10^6$ bites
$= 32 \times 10^6$ bites

throughput = minimum capacity of all these links.

c. repeat (a) & (b) with R_2 reduced to 100 kpps.

=> To calculate the throughput we need to consider the link that has lowest bandwidth along the path.

In this case the lowest bandwidth is R1 which is 500 kbps.

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To calculate the time taken to transfer this file we need to divide the file size by throughput.

$$\text{time} \Rightarrow \frac{\text{file size}}{\text{throughput}}$$

file size $= 4 \times 10^6$ bytes

$= 4 \times 8 \times 10^6$ bits

$= 32 \times 10^6$ bits

$= 32000000$ bits

throughput $= 500$ kbps $\Rightarrow 500 \times 10^3$ b/s

$$\boxed{\text{time taken to transfer the file} = \frac{\text{file size}}{\text{throughput}}}$$

$$= \frac{\cancel{32000000}^{64}}{\cancel{500000}} \frac{\text{bit}}{\text{bit/sec}}$$

$= 64$ sec $\Rightarrow 1$ min 4 sec

1 2 2 5

12 $\times 10^{-1}$ 0.12×10. 0.125×10^{2}

$= 12 \times 10^{-4} \Rightarrow 1.2 \times 10^{-3}$ sec

$= 1.2$ ms //

μ = micro $= 10^{-6}$

m = mili $= 10^{-3}$

M = Mega $= 10^{6}$

K = kilo $= 10^{3}$

c = centi $= 10^{-2}$

An application - layer protocal defines type of message exchanged
eg:- request, response

message syntax: what fields in messages & how fields are delineated
Message semantics :- meaning of info in fields
rules for when & how processes send & respond to messages.
open protocals :- HTTP, SMTP
Proprietary Protocols :- skype (microsoft), zoom
MIME :- anything other than text
Multipurpos mail extention.

google.com + logo bmoce
text + image

Total delay end - to - end delay

Total Delay = $d_{trans} + d_{prop}$

$= 4 + 10 \Rightarrow 14$ msec

The total end-to-end delay (ignoring processing and queuing delay) is the sum of the transmission & propagation delay.

for 16 ms of d_{trans}

Total Delay = 16 + 10 = 26 msec

key.

Does

No, the propagation delay (d/s) is solely based on the physical

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$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

a) $d_{prop} = \frac{\text{distance}}{\text{Speed}} = \frac{m}{s}$ sec

b) $t_{trans} = \frac{L}{R}$ sec

c) $d_{nodal} = \underline{d_{proc}} + \underline{d_{queue}} + d_{trans} + d_{prop}$

$$\boxed{d_{nodal} = d_{trans} + d_{prop}}$$

d end-to-end

d) At $t = 0$ starts
At $t = d_{trans} \Rightarrow \frac{L}{R}$ The last bit just leaving the Host A.

to $\longrightarrow$ L/R
bit bit $\longrightarrow$ bit
1 $\rightarrow$ 1
just started

e) The first bit is in the link and has not reached Host B

$d_{prop} > d_{trans}$ at $t = d_{trans}$

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Consider a network that is propagate in the assignment data in the college network. each student uploads ~~one~~ 1Mb of data with the transmission rate of 1 Mbps calculate the total delay in this network if distance b/w student & faculty machine is 1 km apart with the suitable assumption of propagation delay.

8192 bits
⇒ 1024 bytes

$L = 1Mb = \cancel{1 \times 10^6 \text{ bits}} \Rightarrow \cancel{1024 \text{ bits}}$

$R = 1 Mbps = 1 \times 10^6 \text{ bps}$

$d = 1 km = 1 \times 10^3 \text{ m}$

$S = 2.5 \times 10^8 \text{ m/s}$

$d_{end-to-end} = d_{trans} + d_{prop}$

$d_{trans} = \frac{L_{bits}}{R_{bits/sec}} = \frac{1024 \times 8}{1 \times 10^6} = \cancel{0.001024}$ 8.192

$= \cancel{1.024 \text{ ms}}$

$d_{prop} = \frac{d_m}{S_{m/sec}} = \frac{1 \times 10^3}{2.5 \times 10^8 s} = 0.000004$

$= 0.004 \text{ ms}$

$d_{end-to-end} = \cancel{1.024} \; 8.192 + 0.004 \Rightarrow 8.028 \text{ ms}$